

Mining 3D Data Viewer

A Web-Based 3D Viewer for mXrap Export Files

Business Requirements Analysis

Course	CITS5206 - Information Technology Capstone Project
Group	Group 12
Project	Mining-3D-data-viewer (GitHub repository)
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Revision History:

Version	Date	Author	Description
0.1	06.08.2026	Viktor	Initial draft
0.2	12.08.2026	Viktor	2 nd iteration
0.3	16.08.2026	Lola	3 rd iteration

Table of Contents

1. Problem Statement	3
1.1 Current Problem	3
1.2 Project Objective	3
1.3 Target Users	3
1.4 Long-Term Goal	3
1.5 Primary User Journey	3
1.6 Pain and Gain Map	4
2. Client Communication and MVP Agreement	5
2.1 Communication	5
2.2 MVP Goal	5
2.3 Success Criteria	5
2.4 MVP Deliverables	5
2.5 MVP Feature Prioritisation	5
2.6 Approval	6
3. Project Management and Plans	7
3.1 Project Plan	7
4. Risk and Technology Assessment	8
4.1 Risk Assessment	8
4.2 Skill Assessment	8
4.3 Technology Stack	10
4.3.1 Client-side	10
4.3.2 Server-side	10
4.3.3 3D Visualisation	11
Appendix: Project Resources and Client Communication Records	12
Appendix A: Project Resources	12
Appendix B: Client Communication Records	12
Appendix C: AI Use Acknowledgement	12

1. Problem Statement

1.1 Current Problem

Geotechnical engineers use mXrap to analyse mining data in interactive 3D views. However, the people receiving their reports may not have:

- mXrap installed
- an mXrap licence
- easy access to the original analysis environment

Currently, engineers may need to use screenshots or PowerPoint presentations. These formats are static and do not allow the recipient to rotate the model, inspect spatial relationships, switch data series, or move between different scenes.

1.2 Project Objective

Develop a web-based application that opens an exported mXrap file and displays an interactive 3D visualisation without requiring the client to install mXrap or additional applications.

1.3 Target Users

Primary user: Mine engineers and managers receiving an analysis report for decision-making.

Secondary user: Engineers who prepare and share the exported report.

1.4 Long-Term Goal

Enable primary users to visualise the analysis report without mXrap application through an interactive website.

1.5 Primary User Journey

Step	User action	System response	Expected outcome
Receive export file	The primary user receives the exported file	-	The primary user has the exported file
Open website	The primary user opens website	The browser loads the web application	The website is ready
Select local file	The primary user selects the exported file from their local device	The web application reads the selected file locally in the browser	The file is available for local processing

Validate file	The primary user waits	The web application checks the format and required data	The primary user receives success or error
View scene	The scene opens in the web application	The web application renders the export data	The primary user sees the rendered data
Navigate	The user rotates, zooms and pans camera	The camera position updates in real time	The primary user can inspect the 3D model
Apply filters	The primary user changes visibility of selected data series	The website updates the scene	The primary user reviews only relevant data
Review annotations	The primary user reads labels connected to the point in the 3D model	The labels remain anchored while camera moves	The annotation is displayed in the right position
Switch scenes	The primary user changes scene	The website updates the scene	The primary user reviews the next scene
Finish	The user closes the 3D model or website	The active visualisation session ends and temporary in-memory data is released	Mining data is not uploaded to an external application server.

1.6 Pain and Gain Map

User job	Pain	Gain
Receiving a mining report	The client may not have access to mXrap application	The report can be opened in a web browser
Reviewing 3D scenes across time	Screenshots show only a static view in a fixed point in time	The client can rotate, zoom, and pan the 3D scene across time
Focusing on selected data	Screenshots show static data and may contain too much information	The client can filter data series

2. Client Communication and MVP Agreement

2.1 Communication

Client meeting: Weekly in-person report. Current time slot: Thursdays 4 PM.

Facilitating meetings: Fortnightly. Current time slot: Thursdays 5:30 PM.

Other communications: email and MS Teams.

2.2 MVP Goal

Allow a user to open a local mXrap export file in a browser and view an interactive 3D model containing points, surfaces, annotations and markers.

2.3 Success Criteria

The MVP will be considered successful when:

- a user can open a valid local export file
- the file is processed inside the browser
- the main 3D geometry and point data are displayed
- the user can rotate, zoom and pan the model
- the user can toggle data series
- the user can switch between scenes
- the mining data is not uploaded to an external application server

The preferred architecture – local file loading, browser-memory processing and client-side rendering.

2.4 MVP Deliverables

The Web-based 3D viewer will include:

- Import and validation of an mXrap export file
- Rendering of points, surfaces, markers and annotations
- Camera navigation: rotate, zoom, tilt and pan
- Scene switching
- Data-series filtering
- Client-side processing with no data upload to the web server

2.5 MVP Feature Prioritisation

High impact and high urgency:

- web-based application
- local file import
- client-side file processing
- interactive 3D visualisation
- point and sprite visualisation
- camera rotation, zoom and pan
- scene switching
- display annotations
- marker styles rendering
- data filters

High impact and low urgency:

- time-based data visualisation
- charts and dashboards
- large file rendering

Low impact:

- adding new scenes
- creating and editing annotations
- saving or exporting a modified file

Out of scope for MVP:

- development of mXrap exporter
- development of mXrap API
- development of mXrap plugins
- server-side rendering
- user authentication
- modifying of the original export file
- integration with third-party applications

2.6 Approval

The MVP and prioritisation were reviewed with the client on 13.08.2026. The client confirmed the proposed MVP for the current project stage.

3. Project Management and Plans

3.1 Project Plan

Task	Owner	Assigned to	Deadline
Requirement Analysis Documentation	Viktor	Viktor	17.08.2026
Group Assignment 1	Viktor	Team	17.08.2026
Set up GitHub repository	Shantanu	Shantanu	17.08.2026
Research Three.js or VTK.js for 3D visualisation	Warson	Warson	20.08.2026
Deploy static website (hosting)	Amritansh	Amritansh	24.08.2026
Develop parser for exported file	Amritansh	Amritansh, Hongfei	31.08.2026
Develop validator	Amritansh	Amritansh, Hongfei	04.09.2026
Develop 3D visualiser (core Three.js scene setup)	Hongfei	Hongfei, Amritansh	24.08.2026
Develop camera navigation	Warson	Lola, Warson	31.08.2026
Implement points rendering	Lola	Lola, Warson	31.08.2026
Implement surface rendering	Shantanu	Shantanu, Viktor	07.09.2026
Implement color interpolation	Warson	Warson, Viktor, Lola	07.09.2026
Implement filters	Warson	Lola, Warson	07.09.2026
Implement scenes	Amritansh	Amritansh	14.09.2026
Implement markers	Lola	Lola, Warson	14.09.2026
Implement annotations	Shantanu	Viktor, Shantanu	14.09.2026
Testing	Viktor	Team	26.09.2026
Client demonstration	Viktor	Viktor, Lola	06.10.2026

Project tools:

GitHub – version control and pull requests

MS Teams – team communication

GitHub, MS SharePoint – project documentation

4. Risk and Technology Assessment

4.1 Risk Assessment

Risk	Likelihood	Impact	Mitigation
Export format is insufficiently documented	High	High	Analyse sample export early and confirm the format
Sensitive mining data exposure	Low	High	Process files locally
Corrupted export file	Medium	High	Validate export file structure and data
Large datasets cause poor browser performance	Medium	Medium	Prototype rendering early and define max size of export file
Browser compatibility	Low	Medium	Define supported browsers
Failure to meet project deadlines	Medium	Medium	Weekly sprint meetings, assign clear responsibilities and use project management tools and identify possible delays
Failure to meet the agreed MVP scope (scope creep)	Low	Medium	Define MVP features early, review progress against the scope weekly and prioritise features based on importance and time
Very large export files (client indicated up to ~100 GB) may exceed browser memory capacity during client-side parsing	Medium	High	Investigate chunked/streaming parsing and selective (lazy) loading of scene data rather than loading the full file into memory at once; define a maximum supported file size for the MVP and communicate this limitation to the client early.

4.2 Skill Assessment

Skill	Capability	Gap	Action
JavaScript	Good	Low	-
Python	Good	Low	-

React / Next.js	Moderate	Medium	Amritansh has direct prior experience; pair programming and shared code reviews to spread knowledge across the team.
Django	Moderate	Medium	-
GitHub	Good	Low	-
Three.js (3D rendering)	Limited	High	Research documentation and tutorials; build small proof-of-concept prototypes early (see Section 4.3.4).
Browser File API / large-file handling	Limited	High	Investigate chunked/streaming file reading early; prototype with large sample export files before committing to an approach (see Section 4.1).
Data parsing (export file formats)	Moderate	Medium	Analyse sample export files early to confirm structure and required fields.
Performance optimisation (large datasets)	Limited	Medium	Prototype rendering early; profile performance and optimise iteratively; define a maximum supported file size for the MVP.
UI/UX	Moderate	Low	Design review.
Testing	Good	Low	-
Cybersecurity	Moderate	Medium	Review secure local file handling and implement basic protective measures.

4.3 Technology Stack

4.3.1 Client-side

Frontend

React with Next.js will be used for the web interface, with the application configured for client-side execution and static deployment (consistent with the architecture described in Section 4.3.1). This technology stack was proposed to the client during the meeting on 13 August 2026.

The primary rationale is team familiarity and implementation risk reduction. Several team members have existing JavaScript and React experience, and Amritansh has previously developed a web application using Django and Next.js. Next.js also provides a mature and well-documented React development ecosystem. Given the team's limited prior experience with browser-based 3D visualisation, selecting a familiar web development framework allows more development effort to be focused on file processing and 3D visualisation rather than on learning an additional application framework.

4.3.2 Server-side

Backend

No dedicated backend application framework is required for the MVP. The client explicitly requested a client-side architecture in which exported mXrap data is loaded, parsed, and rendered locally within the user's browser, ensuring that mining data is not transmitted to an external application server. This requirement was discussed during the initial requirements meeting and reaffirmed by the client on 13 August 2026. The only server-side requirement is therefore static hosting to deliver the compiled web application files to the browser.

Alternative considered: A Django backend was informally discussed by the team before the architecture requirements were confirmed with the client, particularly in case server-side processing of large export files was required. This approach was ruled out because the client indicated that export files may reach approximately 100 GB. Uploading files of this size to an application server would be impractical and would conflict with the client's requirement that potentially sensitive mining data remain local. It would also introduce unnecessary data-privacy and cybersecurity risks. Efficient handling of very large files on the client side therefore remains an important implementation consideration for the MVP (see Section 4.1).

Database

Not required. No application database or server-side persistent storage is required for the MVP. Mining data will be processed locally within the user's browser and will not be persisted to an external server.

This design is consistent with the client's confirmed requirement that mining data must not be uploaded to an external application server. Any temporary data required for parsing and visualisation will remain on the client side for the duration necessary to support the active visualisation session; the specific in-browser storage mechanism (e.g. in-memory objects, File API, or IndexedDB for caching) is an implementation detail to be determined during development.

4.3.3 3D Visualisation

Two primary libraries were considered for browser-based 3D visualisation: Three.js and VTK.js.

Three.js was recommended by the project facilitator, Fudong Qin. He also noted on 14 August 2026 that Three.js is used for a comparable mining-related 3D visualisation project within his own organisation, providing relevant real-world validation for its use in this domain.

VTK.js was suggested by the client because the existing mXrap desktop application is built using the C/C++ version of VTK, which could provide conceptual alignment with the existing mXrap visualisation architecture and its marker/colour-ramp behaviour. However, the client noted that they do not have direct experience with VTK.js and expressed no strong preference between the two libraries, encouraging the team to select whichever option proves more practical to implement.

Decision: Three.js. Neither library has previously been used by the team; however, several members already have JavaScript and React experience, and Three.js provides extensive JavaScript-native documentation, examples, and community resources, which fits directly with the React/Next.js stack chosen in Section 4.3.2. Combined with the facilitator's recommendation and evidence of its use in a comparable mining visualisation project, Three.js currently presents the lower implementation risk for the team within the MVP timeframe. VTK.js remains a technically relevant alternative, given its conceptual alignment with mXrap's own visualisation engine, and may be revisited should later prototyping reveal requirements that are better supported by the VTK ecosystem.

Appendix: Project Resources and Client Communication Records

Appendix A: Project Resources

GitHub Repository: <https://github.com/sharmashantanu319/Mining-3d-data-viewer>

MS Teams Channel: [Group 12 | CITS5206 - Information Technology Capstone Project SEM-2 2026](#)

Appendix B: Client Communication Records

All client and facilitator meetings are recorded and transcribed; recordings/transcripts are retained by the team as supporting files.

Client Meeting 1 – Initial Requirements 30 July 2026: Initial requirements gathering with the client, covering the problem statement, target users, architecture preference (client-side, no server upload), and data size considerations.

Facilitator Meeting – 5 August 2026: Team formation and informal technical discussion on client-side architecture and initial technology ideas.

Client Meeting 2 – 13 August 2026: Requirements and MVP prioritisation review, technology stack discussion (Three.js vs VTK.js, frontend/backend options), rendering and scenes clarification, export file structure, and annotation format questions.

Appendix C: AI Use Acknowledgement

AI was used to assist with preparing Minutes of Meetings by summarising and structuring meeting notes. All AI-generated content was reviewed and adjusted for accuracy.